APPROVED OPERATIONAL RUNBOOK: FXT-RUN-004

SWIFT / Fedwire Payment Gateway 504 Gateway Timeout Recovery Runbook

Application: Global Payments & Swift Settlement (FXT-Pay)	**Service:** Swift-Alliance-Gateway-HTTP	**Region:** GLOBAL / AMER / EMEA
Environment: PROD	**Category:** External Banking & Payments API	**Version:** 4.1.0
Owner / Team: Payments & Clearing Support Pod	**Last Updated:** 2026-06-19	**Doc ID:** FXT-RUN-004

1. Operational Symptoms & Issue Summary

Issue Summary: Outbound cross-border wire transfers and MT103 / pacs.008 payment messages timing out with HTTP 504 Gateway Timeout or TCP handshake reset at the perimeter secure gateway.

Detectable Symptoms & Alerts:

- HTTP Response Code: '504 Gateway Timeout from https://swift-gateway.fxtrading.internal/api/v2/wires'
- Error: 'SSLHandshakeException: PKIX path validation failed or cert expired'
- Alert: 'SWIFT_OUTBOUND_BUFFER_SPIKE: 420 pending wire messages queued without ACK'
- Business impact: Risk of missing central bank intraday cutoff (Fedwire 17:00 EST / Target2 18:00 CET)

2. Root Causes & Technical Diagnosis

- Perimeter Mutual TLS (mTLS) client certificate expiration on API Gateway proxy.
- Upstream SWIFT Alliance Access interface maintenance without prior notice.
- Hardware Security Module (HSM) signature cluster latency exceeding 8,000ms.

3. Mandatory Safety Directives (DO NOT DO)

CRITICAL OPERATIONAL WARNING: **DO NOT resubmit payment requests with new UUIDs if the gateway response is unknown or timed out; doing so creates DUPLICATE WIRE TRANSFERS.**
CRITICAL OPERATIONAL WARNING: **DO NOT bypass HSM digital signature verification under any circumstance; violates SWIFT CSP compliance.**
CRITICAL OPERATIONAL WARNING: **NEVER disable TLS certificate validation flags (--insecure or -k) in production payment scripts.**

4. Step-by-Step Resolution Procedures

Step 1: Validate Gateway HTTP Connectivity and TLS Certificate Expiry

Test TLS handshake against internal proxy endpoint and verify certificate validity dates.

Command (PowerShell):

```
$cert = [System.Net.ServicePointManager]::ServerCertificateValidationCallback;
Test-NetConnection -ComputerName swift-gateway.fxtrading.internal -Port 8443
```

Step 2: Query SWIFT Acknowledgment State for In-Flight Wires

Check database for in-doubt payment IDs before initiating retry or failover.

Command (SQL):

```
SELECT payment_id, currency, amount, status, swift_uetr, created_time FROM t_payment_wire WHERE
status IN ('SENT_PENDING_ACK', 'TIMEOUT') AND created_time > SYSDATE - 1/24;
```

Step 3: Switch Payment Route to Standby Secondary Node

Trigger approved automated route switch to Secondary Swift Gateway instance B.

Command (PowerShell):

```
Invoke-RestMethod -Uri 'https://config-svc.fxtrading.internal/api/routes/swift' -Method POST
-Body '{"active_gateway": "SWIFT_GATEWAY_NODE_B"}' -ContentType 'application/json'
```

Step 4: Reconcile In-Doubt Wires via Status Query API

Poll SWIFT UETR tracking status for all unacknowledged wires.

Command (Bash/Shell):

```
curl -X POST https://swift-gateway-b.fxtrading.internal/api/v2/reconcile-uetr -H 'Content-Type:
application/json' -d '{"batch_window": "CURRENT_HOUR"}'
```

5. Post-Recovery Verification Procedures

V-1: Confirm HTTP response 200 OK received for gateway health check endpoint /actuator/health.

V-2: Verify all in-flight payments change status from 'SENT_PENDING_ACK' to 'CONFIRMED_ACK_RECEIVED'.

V-3: Zero new 504 errors in Grafana Payment Gateway Dashboard for 15 consecutive minutes.

6. Dependency-Aware Service Recovery Sequence

Ordered Sequence:
1. HSM Security Token Service Health Check
   ➔ 2. Secondary SWIFT Gateway Node B Health Verification
   ➔ 3. API Route Switch to Node B
   ➔ 4. In-Doubt UETR Reconciliation Run
   ➔ 5. Resume Payment Dispatcher Queue

Explicit Stop/Escalate Gate: If Node B also returns 504, STOP immediately. Do NOT dispatch queued wires. Escalate to Head of Treasury Ops and Network Security.

7. Edge Cases & Escalation Protocol

• **Edge Case:** Wire amount exceeds single transaction limit ($50,000,000): Requires manual 4-eyes approval in Payment Portal.

- **Edge Case:** OFAC / Sanctions screening false positive: Hold transaction in PENDING_COMPLIANCE queue.

Primary Contact: Treasury & Payments Ops On-Call (payments-ops-oncall@fxtrading.internal | Phone: ext. 8831)	**Secondary / Bridge:** SWIFT Alliance Technical Lead (swift-tech@fxtrading.internal)
Severity / SLA: P1 Critical: 5-minute response SLA due to cut-off penalty	**Escalation Condition:** Failure to route within 15 minutes of central bank cutoff deadline.